ASYNCHRONOUS STATE RE-ANCHORING HANDSHAKE UNDER SAFETY INTERVENTION

Surviving real-time MCU safety vetoes: atomic mailbox notification, queue purge, KV-cache reset, and bumpless restart

1. MCU SAFETY VETO (t = 120ms)

CBF Invariant Breached

$\dot{h}(x) + \gamma h(x) < 0$

Overrides proposed action

Dynamic obstacle encroaches

Executes Category 2 Stop

- Latches physical state:

`q_actual = [1.24, -0.42, ...]`

`q̇_actual = [0.0, 0.0, ...]`

2. SHARED SRAM POSTING

INTER-CORE MAILBOX

ATOMIC VETO PULSE

`IPC_Hardware_Mailbox`

`veto_flag = 1;`

Microsecond interrupt latency

`epoch_seq = 4092;`

- Notifies Linux MPU immediately.
- Zero dynamic memory allocation.

3. MPU PURGE & RESET

1. Flush Action Queue

2. Clear KV-Cache

- Purges all stale chunk waypoints
- Resets Transformer autoregression
- Aborts active trajectory decoders
- Eliminates open-loop state hallucination

- Ingests `q_actual` as new initial boundary condition *q(0)*.

4. BUMPLESS RESUMPTION

NEW TRAJECTORY SEED

```
q_start = q_actual;
q̇_start = q̇_actual;
C^2 Quintic Fit
BUMPLESS RESUME
```

- Zero acceleration jump.
- Resynthesizes safe chunk.
- Motor resumes smooth motion.

THE PHYSICAL AI STATE RE-ANCHORING CONTRACT

1. **The Hallucination Danger:** If an MPU planner generates trajectories assuming an un-vetoed path, its next chunk commands a massive step jump from the actual physical stopping point.
2. **Strict Hardware Handshake:** By enforcing that every new chunk starts at the exact measured physical state (*q_actual, q̇_actual*), position error is zeroed.
3. **Governed Recovery:** The system gracefully recovers from dynamic obstacles without rebooting the robot or entering an unrecoverable fault state.